PART 10

Data Safety & Resilience

81 Database Migrations

82 Schema Versioning

83 Disaster Recovery

84 Backups

85 Failover

86 Multi-Region Deployment

87 Chaos Engineering

#081

Database Migrations

WHAT IT IS

Versioned, repeatable scripts that change the DB schema.

WHY IT MATTERS

Schema changes travel with code across all environments.

EXAMPLE

```
Schema::table('tasks', function (Blueprint $t) {
    $t->timestamp('archived_at')->nullable()->after(
    'status');
});
```

#082

Schema Versioning

WHAT IT IS

Tracking the schema's version and keeping app and DB compatible over time.

WHY IT MATTERS

Rolling deploys run old and new code at once.

EXAMPLE

Laravel's migrations table records applied migrations. For events/APIs/queues, include a version: {"type":"task.created","v":2,...} and support the previous version during transition. Never rename a column in one step; add the new one, dual-write, switch reads, drop the old.

#083

Disaster Recovery

WHAT IT IS

The plan and tooling to restore service after a major failure (region outage, corruption, ransomware).

WHY IT MATTERS

Defines how long you are down and how much data you lose.

EXAMPLE

Define RTO (max downtime, e.g. 1h) and RPO (max data loss, e.g. 5 min). Write a runbook, store backups in another region/account, and practice restores quarterly. An untested backup is a hope, not a backup.

#084

Backups

WHAT IT IS

Copies of data that can be restored later.

WHY IT MATTERS

Protects against deletion, bugs, corruption and attacks; replication alone does not (it copies mistakes).

EXAMPLE

```
pg_dump -Fc -h db -U app taskflow | gzip >
    taskflow_$(date +%F).dump.gz
aws s3 cp taskflow_*.dump.gz s3://backups-eu/ --sse
```

#085

Failover

WHAT IT IS

Automatically switching to a standby when the primary fails.

WHY IT MATTERS

Cuts downtime from hours to seconds.

EXAMPLE

Patroni promotes a Postgres replica; Redis Sentinel promotes a replica; load balancer removes unhealthy app nodes. Laravel reconnects if you handle exceptions and keep timeouts short. Beware split-brain: ensure the old primary is fenced.

#086

Multi-Region Deployment

WHAT IT IS

Running the system in multiple geographic regions.

WHY IT MATTERS

Survives regional outages and gives users lower latency.

EXAMPLE

Active-passive (standby region takes over via DNS/Global Load Balancer) is simpler. Active-active needs data replication and conflict resolution (CAP trade-offs). Route users with GeoDNS; keep data residency laws in mind.

#087

Chaos Engineering

WHAT IT IS

Deliberately injecting failures in a controlled way to test resilience.

WHY IT MATTERS

Finds weaknesses before real incidents do.

EXAMPLE

```
kubectl delete pod -l app=taskflow-web
    --field-selector=status.phase=Running
    --wait=false
```

Found this useful?

Save it for later.

Share it with a teammate.

Full guide with all code examples:
the Backend Engineering Handbook (GitHub).